Alex Raymond Hyman

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EDUCATION

Informatics AIAI, University of Edinburgh Sept. 2025 – Present
Neurosymbolic AI & Formal Verification PhD Student Edinburgh, Scotland

- **Formally verified** an imperative reverse **automatic differentiation engine** in Isabelle/HOL using separation logic.
- Developed formal proofs for a **core backpropagation algorithm** to ensure reliable gradients in safety-critical AI systems.
- **Lead author** on peer-reviewed paper detailing the verification methodology; **accepted and presented** at the Workshop on Logical Methods for Neural Network Analysis (LogicNN) at **FLoC, Lisbon** 2026 (awaiting publication).
- **Course creator** for **new NeuroSymbolic AI course** at University of Edinburgh
- Attended **prestigious Marktoberdorf Summer School** on Engineering Secure and Dependable Software Systems
- Well-versed in **agentic theorem proving setups** with **Claude Code/Codex/OpenCode**
- Believe that proof assistants (**Lean & Isabelle**) with AI is ushering in a new era of verified large-scale software

University of Edinburgh 2020 – 2025
Masters in Mathematics (MMath) Edinburgh, Scotland

- **Prizes**: Statistics and Probability Prize (Top Score) & Distinguished Overall Honours Exams Performance.
- **Courses**: Machine Learning, Automated Theorem Proving, Algorithms, Computer Systems, Real/Complex Analysis, Axiomatic Set Theory, Quantum Mechanics, Financial Mathematics, etc.
- **Dissertation**: Formalising Neural Networks in an Interactive Theorem Prover (86%, Distinction). Engineered formal proofs for deep neural network architectures, establishing a foundational framework for verifiable AI model outputs.

University of British Columbia 2023 – 2024
Mathematics (Year Abroad) Vancouver, Canada

- **Courses**: Metric Spaces, Numerical Linear Algebra, Probability, Green's Functions and Variational Methods

Hutchesons' Grammar School 2014–2020
High School Glasgow, Scotland

- Advanced Higher 3A's: Physics, Mechanics and Pure Mathematics. Higher 7A's incl. Computer Science & Economics.

EXPERIENCE

AI Modelling Lab, University of Edinburgh June 2023 – Sept. 2023
Summer Research Internship Edinburgh, Scotland

- Introduced to interactive theorem provers and implemented neural network library in Isabelle/HOL.

Hymans Robertson Aug. 2019 – Present
Actuarial Trainee (Internship) Glasgow, Scotland

- Executed technical actuarial calculations and aggregated Task Force on Climate-Related Financial Disclosures (TCFD) metrics to assess fund environmental impacts.
- Programmed VBA macros to automate complex Excel workflows, reducing manual processing time for routine tasks.

Content Creation Present
Mathematics and Computer Science

- **YouTube (160K views)** and TikTok discussing formal mathematics, philosophy and computer science.

Gordon and Noble, Sheriff Officers June 2021 – Dec. 2021
Office Administrator and Programmer Glasgow, Scotland

- Programmed and sold custom Python automation scripts to replace manual data entry tasks in Microsoft Excel.
- Streamlined the construction of legal documents for debt collection purposes.

SKILLS

Languages: Python, Rust, Haskell, Isabelle/HOL, Lean, VBA, JavaScript
Frameworks & Tools: PyTorch, TensorFlow, Pandas, Git
Hobbies: Climbing, Slacklining, Juggling & Skiing. Cryptic Crosswords & Backgammon.